

6	(a) (i) to concentrate the (magnetic) flux / reduce flux losses	B1	[1]
	(ii) changing flux (in core) induces current in core currents in core give rise to a heating effect	M1 A1	[2]
	(b) (i) e.m.f. induced proportional to rate of change of (magnetic) flux (linkage)	M1 A1	[2]
	(ii) magnetic flux in phase with / proportional to e.m.f. / current in primary coil e.m.f. / p.d. across secondary proportional to rate of change of flux so e.m.f. of supply not in phase with p.d. across secondary	M1 M1 A0	[2]
	(c) (i) for same power (transmission), high voltage with low current with low current, less energy losses in transmission cables	B1 B1	[2]
	(ii) voltage is easily / efficiently changed	B1	[1]

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7 (a) for a wave, electron can 'collect' energy continuously

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